

Liang 1997 SHE–BTE Reproduction

Richardson / Grid-Convergence Audit of the drain G_{ii} trio

2026-09-26

Provenance

This audit uses the archived controlled-convergence trio of 2026-09-24 (`runs/2026-09-24_103849`, nodes `data/she2d_richardson_{71x58,100x70,141x85}.npz`). Those files were **produced by the pre-freeze solver revision d30b471**; they were subsequently validated as **numerically equivalent to the frozen solver solver-frozen-2026-09-26** within the solver’s own run-to-run noise ($\sim 4 \times 10^{-6}$ relative on G_{ii}) and far inside the 3×10^{-4} certification gate (40×34 port regression: G_{ii} rel 4.35×10^{-6}). They are *not* relabelled as generated by the frozen tag.

Common invariants across all three nodes (verified, fail-closed): mesh law (ratio 6.0, $W = 0.09$, $X_C = 0.60 \mu\text{m}$), $dH = 0.0125 \text{ eV}$, $\varepsilon_{\text{max}} = 3.02 \text{ eV}$, $N_H = 591$, `she_tol` = 10^{-8} , common frozen H -span, H-grid SHA1 `b4319bb03303`. Bias $V_g = V_d = 3 \text{ V}$, $\Phi_{\text{gate}} = -0.740 \text{ V}$. Each node: fresh matrix + fresh split-ILU + cold Krylov every outer iteration.

1. Per-node certification

All three nodes reached the *unchanged* certified plateau gate (last-4 cumulative spread of G_{ii} and $G_{\text{tot}} \leq 3 \times 10^{-4}$, $R_\phi \leq 1.5 \times 10^{-3}$).

node	$h_{\text{rep}} (\mu\text{m})$	$G_{ii}^{\text{max}} (\text{cm}^{-3}\text{s}^{-1})$	$G_{\text{tot}} (\text{cm}^{-1}\text{s}^{-1})$	$T_e^{\text{max}} (\text{K})$	$R_\phi (\text{V})$	bal	iters
71×58	1.356×10^{-3}	3.0756×10^{25}	9.265×10^{13}	3184	3.9×10^{-5}	$\sim 1.7 \times 10^{-4}$	359
100×70	0.960×10^{-3}	2.1362×10^{25}	6.025×10^{13}	3161	2.6×10^{-4}	3.9×10^{-5}	480
141×85	0.676×10^{-3}	2.3874×10^{25}	6.526×10^{13}	3194	7.0×10^{-4}	6.6×10^{-5}	60 [†]

[†]141×85 warm-started from the converged 100×70, hence 1 chunk. All gate-PASS; SHE linear residual $\sim 10^{-8}$; number balance |bal| $\lesssim 2 \times 10^{-4}$. Each node is individually a converged, conservation-respecting fixed point.

2. Grid-to-grid behaviour — NON-MONOTONE

Refining h_{rep} by $\sqrt{2}$ per step (ratios 1.412, 1.420), the drain peak G_{ii} does *not* move in one direction:

$$\underbrace{3.0756}_{71 \times 58} \xrightarrow{-30.5\%} \underbrace{2.1362}_{100 \times 70} \xrightarrow{+11.8\%} \underbrace{2.3874}_{141 \times 85} \quad [\times 10^{25} \text{ cm}^{-3}\text{s}^{-1}].$$

The sequence reverses direction (down, then up). G_{tot} behaves the same ($9.265 \rightarrow 6.025 \rightarrow 6.526 \times 10^{13}$; -35% then $+8.3\%$). T_e^{max} is essentially flat (3184/3161/3194 K, $\pm \sim 20 \text{ K}$). **This reversal is the central fact of the audit.**

3. Is Richardson extrapolation even valid here?

Geometry check (passes): $r_x = 1.414$, $r_y = 1.410/1.425$, both within 5% of $\sqrt{2}$, so a single representative $h \equiv h_{\text{rep}}$ is defensible (spacings taken from the *physical* mesh, not from N_x).

Asymptotic-model check (fails): classical Richardson assumes a one-term regime $Q(h) = Q^* + C h^p + \dots$, for which the observed order follows from

$$r^p = \frac{Q_1 - Q_2}{Q_2 - Q_3} = \frac{+9.40 \times 10^{24}}{-2.51 \times 10^{24}} = -3.74.$$

A *negative* ratio has **no real solution** p ($r^p > 0$ for every real p): the three nodes are not describable by a single-term power law. The gated fit flags the reversal as *genuine*, not solver noise:

$$S_{\text{rev}}(G_{ii}) = \frac{|Q_2 - Q_3|}{u_{23}} = \frac{2.51 \times 10^{24}}{3.41 \times 10^{21}} = 737 \gg 1, \quad S_{\text{rev}}(G_{\text{tot}}) = 524 \gg 1,$$

where $u_{23} \sim 3 \times 10^{21}$ is the per-node plateau/solver uncertainty (five orders of magnitude below the reversal). **Three-node Richardson extrapolation is therefore not justified for G_{ii} (nor G_{tot}) on this mesh family, and the harness suppresses it.**

4. Extrapolated quantities: NOT reported

Because the asymptotic-model check fails, we deliberately do *not* report an observed order p_{obs} , a continuum G_{ii}^∞ , or a GCI/discretisation-error band — manufacturing an extrapolated value from a sign-reversing sequence would be numerically meaningless. The correct audit result is the *absence* of a justified extrapolation. The tested meshes bound $G_{ii}^{\text{max}} \in [2.14, 3.08] \times 10^{25}$ with finest-grid value 2.39×10^{25} ; no continuum value is claimed. A rigorous continuum estimate would require a second mesh family (fixed y -mesh, refine drain- x) to disentangle the anisotropic x/y error and re-test for a consistent p .

5. Scientific verdict — two separate questions

- (a) **Absolute G_{ii} convergence: NOT demonstrably asymptotic.** The three drain-refined nodes do not occupy a single asymptotic regime; no continuum peak G_{ii} can be rigorously quoted. The absolute drain-avalanche magnitude remains grid-limited.
- (b) **Qualitative / physical conclusions: ROBUST.** The non-monotone reversal itself is a robust finding ($S_{\text{rev}} \gg 1$, far above noise), not a numerical artefact. The broader reproduced physics is likewise robust and independent of this extrapolation gap: inversion/transport ($n^{\text{max}} \approx 1.0 \times 10^{20}$, $|v| \approx 1 \times 10^7$ cm/s, $T_e^{\text{max}} \approx 3180$ K), drain-localised avalanche, and the $\sim 37\%$ self-consistent G_{ii} -feedback suppression established across the coupled runs.

These must not be conflated: the absolute avalanche magnitude is not asymptotically converged, *yet* the qualitative picture (drain-localised, self-consistently suppressed, non-monotone under refinement) stands. That is the honest, and scientifically sufficient, closeout of the reproduction.

Audit of the 2026-09-24 archived trio (pre-freeze d30b471; validated equivalent to solver-frozen-2026-09-26). Gated fit: `src/richardson_fit.py`. Certified gate unchanged (3×10^{-4} spread, $R_\phi \leq 1.5 \times 10^{-3}$).